```java
import java.util.Scanner;
public class App {

    // Global variables
    private static int[][] planeSeats = null;
    private static int[] pricePerRow = null;

    public static void main(String[] args) {
        System.out.println("Welcome to Flying Java!");
        initialiseRows();
        runMenu();
    }

    public static void initialiseRows() {
        planeSeats = new int[4][]; // total rows - multidimensional array
        planeSeats[0] = new int[18]; // row 1 - initialised at 0 all available
        planeSeats[1] = new int[20]; // row 2 - initialised at 0 all available
        planeSeats[2] = new int[20]; // row 2 - initialised at 0 all available
        planeSeats[3] = new int[18]; // row 2 - initialised at 0 all available
        pricePerRow = new int[4];
        pricePerRow[0] = 50;
        pricePerRow[1] = 80;
        pricePerRow[2] = 80;
        pricePerRow[3] = 50;
    }

    public static void runMenu() {
        int option;
        boolean cont = true;

```

```
31          while (cont) {
32              option = getOption();
33              switch (option) {
34                  case 0:
35                      cont = false;
36                      break;
37                  case 1:
38                      buyTicket();
39                      break;
40                  case 2:
41                      showSeatingArea();
42                      break;
43                  default:
44                      System.out.println("Option not available. Please select a vali
45              }
46          }
47      }
48
49      private static int getOption() {
50
51          Scanner input = new Scanner(System.in);
52          boolean valid = false;
53          int option = -1;
54          do {
55              System.out.println();
56              System.out.println("+-----------------------------------------+");
57              System.out.println("|                MAIN MENU                |");
58              System.out.println("+-----------------------------------------+");
59              System.out.println("|  1) Buy a plane ticket                  |");
60              System.out.println("|  2) Show seating area and available seats   |");
```

```
61              System.out.println("|  0) Quit                                                |");
62              System.out.println("+-------------------------------------------------------+");
63              System.out.print("Please select an option: ");
64              try {
65                  option = input.nextInt();
66                  valid = true;
67              } catch (Exception e) {
68                  System.out.println("This option not valid.");
69              }
70          } while (!valid);
71          return option;
72
73      }
74
75      private static void buyTicket() {
76          Scanner input = new Scanner(System.in);
77          System.out.print("Enter row number: ");
78          int row = input.nextInt() - 1;
79          System.out.print("Enter seat number: ");
80          int seat = input.nextInt() - 1;
81
82          // Check if the seat is available or not
83          if (planeSeats[row][seat] == 0) {
84              planeSeats[row][seat] = 1;
85              System.out.println("Purchase successful.");
86              showSeatingArea();
87          } else {
88              System.out.println("This seat is already taken.");
89          }
90
```

```
 91     }
 92
 93     private static void showSeatingArea() {
 94
 95         int rows = planeSeats.length;
 96         char aisle = '|';
 97
 98         System.out.println("=".repeat(76));
 99         System.out.println("                                    PLANE SEATING MAP ");
100         System.out.println("=".repeat(76));
101
102         for (int row = 0; row < rows; row++) {
103             System.out.print("Row " + (row+1) + "(£" + pricePerRow[row]+ ")  ");
104             int seatsPerRow = planeSeats[row].length;
105             for (int seat = 0; seat < seatsPerRow; seat++) {
106                 if (seat == 9) { // Create aisles
107                     System.out.print(" " + aisle + " ");
108                 }
109                 if (planeSeats[row][seat] == 0) { //available
110                     System.out.print("[O]");
111                 } else { // not available
112                     System.out.print("[X]");
113                 }
114             }
115             System.out.println();
116         }
117         System.out.println("=".repeat(76));
118         System.out.println("LEGEND: [O] = Seat available, [X] = Seat not availabl
119         System.out.println("=".repeat(76));
120         System.out.println();
```

```
121
122         }
123 }
124
```